

Job Applicant

Challenge Scenario

Wowza Grand College opened a job application portal where applicants could upload resumes for HR review. Later, the SOC team noticed suspicious certificate enrollments in their AD CS environment and collected evidence from the compromised domain controller.

Evidence Used

- IIS access log: `log.txt`
- Windows Security log export: `security.json`
- NTFS artifacts: `$MFT / MFT.csv` and `$Extend\$.J`
- Screenshots included with this writeup

Research Approach

I did not know every Windows or AD CS artifact upfront, so I treated the investigation as a loop: find a clue in the logs, research what that clue means, then come back to the evidence to test the idea. For web behavior, I checked IIS/ASP.NET handler documentation. For the certificate activity, I researched AD CS audit event IDs before tying them back to `security.json`. For Kerberos and credential dumping, I used MITRE ATT&CK and Microsoft references to make sure the technique names matched the observed events.

Task 1: When did the threat actor first interact with the College's IT infrastructure?

Answer: `2025-07-27 19:27:35`

I started with the IIS log because the case begins with a resume upload website. The first clearly suspicious external request came from `143.198.231.177`:

```
2025-07-27 19:27:35 192.168.189.150 GET /index.aspx - 80 -  
143.198.231.177 ... 404
```

Even though the server returned `404`, this was still the first observed contact from the attacker-controlled IP. That made `2025-07-27 19:27:35` the starting point of the attack timeline.

```
#Software: Microsoft Internet Information Services 10.0
#Versions: 1.0
#Date: 2025-07-27 17:39:51
#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status time-taken
2025-07-27 17:39:51 192.168.189.150 GET / - 80 - 192.168.189.75 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/138.0.0.0+Safari/537.36 - 200 0 0 688
2025-07-27 17:48:07 192.168.189.150 POST / - 80 - 192.168.189.75 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/138.0.0.0+Safari/537.36 http://resumeupload.wowza.edu/ 200 0 0 51
2025-07-27 17:49:46 192.168.189.150 POST / - 80 - 192.168.189.75 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/138.0.0.0+Safari/537.36 http://resumeupload.wowza.edu/ 200 0 0 85
#Software: Microsoft Internet Information Services 10.0
#Versions: 1.0
#Date: 2025-07-27 19:27:35
#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status time-taken
2025-07-27 19:27:35 192.168.189.150 GET /index.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 404 0 0 737
2025-07-27 19:27:42 192.168.189.150 GET /main.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 404 0 0 4
2025-07-27 19:28:18 192.168.189.150 POST /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 200 0 0 57
2025-07-27 19:28:29 192.168.189.150 POST /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/default.aspx 200 0 0 180
2025-07-27 19:28:50 192.168.189.150 POST /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/default.aspx 200 0 0 9
2025-07-27 19:28:50 192.168.189.150 GET /uploads - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 404 0 2 4
2025-07-27 19:28:57 192.168.189.150 GET /robots.txt - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 3
2025-07-27 19:28:57 192.168.189.150 GET / - 80 - 143.198.231.177 feroxbuster/2.11.0 - 200 0 0 2
2025-07-27 19:28:57 192.168.189.150 GET /.htaccess19fbb802049e40d78a338d25ebd13d07 - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 2
2025-07-27 19:28:57 192.168.189.150 GET /cgi-bin - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 3
2025-07-27 19:28:57 192.168.189.150 GET /category - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 12
2025-07-27 19:28:57 192.168.189.150 GET /forum - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 6
2025-07-27 19:28:57 192.168.189.150 GET /modules - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 4
2025-07-27 19:28:57 192.168.189.150 GET /cache - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 7
2025-07-27 19:28:57 192.168.189.150 GET /tmp - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 7
2025-07-27 19:28:57 192.168.189.150 GET /node - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 16
2025-07-27 19:28:57 192.168.189.150 GET /scripts - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 19
2025-07-27 19:28:57 192.168.189.150 GET /stats - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 20
```

Task 2: What is the malicious IP address used for initial reconnaissance and discovery?

Answer: 143.198.231.177

After finding the first request, I followed the same **c-ip** value in the IIS log. The address **143.198.231.177** kept requesting common discovery paths such as **/admin**, **/login**, **/wp-content**, **/uploads**, **/robots.txt**, and many random-looking paths.

The user agent also changed to **feroxbuster/2.11.0**, which made the behavior look like directory brute forcing rather than normal browsing.

```
#Software: Microsoft Internet Information Services 10.0
#Versions: 1.0
#Date: 2025-07-27 17:39:51
#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status time-taken
2025-07-27 17:39:51 192.168.189.150 GET / - 80 - 192.168.189.75 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/138.0.0.0+Safari/537.36 - 200 0 0 688
2025-07-27 17:48:07 192.168.189.150 POST / - 80 - 192.168.189.75 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/138.0.0.0+Safari/537.36 http://resumeupload.wowza.edu/ 200 0 0 51
2025-07-27 17:49:46 192.168.189.150 POST / - 80 - 192.168.189.75 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/138.0.0.0+Safari/537.36 http://resumeupload.wowza.edu/ 200 0 0 85
#Software: Microsoft Internet Information Services 10.0
#Versions: 1.0
#Date: 2025-07-27 19:27:35
#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status time-taken
2025-07-27 19:27:35 192.168.189.150 GET /index.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 404 0 0 737
2025-07-27 19:27:42 192.168.189.150 GET /main.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 404 0 0 4
2025-07-27 19:28:18 192.168.189.150 POST /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 200 0 0 57
2025-07-27 19:28:29 192.168.189.150 POST /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/default.aspx 200 0 0 180
2025-07-27 19:28:50 192.168.189.150 POST /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/default.aspx 200 0 0 9
2025-07-27 19:28:50 192.168.189.150 GET /uploads - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 404 0 2 4
2025-07-27 19:28:57 192.168.189.150 GET /robots.txt - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 3
2025-07-27 19:28:57 192.168.189.150 GET / - 80 - 143.198.231.177 feroxbuster/2.11.0 - 200 0 0 2
2025-07-27 19:28:57 192.168.189.150 GET /.htaccess19fbb802049e40d78a338d25ebd13d07 - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 2
2025-07-27 19:28:57 192.168.189.150 GET /cgi-bin - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 3
2025-07-27 19:28:57 192.168.189.150 GET /category - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 12
2025-07-27 19:28:57 192.168.189.150 GET /forum - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 6
2025-07-27 19:28:57 192.168.189.150 GET /modules - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 4
2025-07-27 19:28:57 192.168.189.150 GET /cache - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 7
2025-07-27 19:28:57 192.168.189.150 GET /tmp - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 7
2025-07-27 19:28:57 192.168.189.150 GET /node - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 16
2025-07-27 19:28:57 192.168.189.150 GET /scripts - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 19
2025-07-27 19:28:57 192.168.189.150 GET /stats - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 20
2025-07-27 19:28:57 192.168.189.150 GET /test - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 19
```

Task 3: The threat actor uploaded a file using the file upload functionality. Under what name did the server store the uploaded resume file?

Answer: Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf

The attacker submitted data to the upload page:

```
2025-07-27 19:32:28 ... POST /default.aspx ... 143.198.231.177 ... 200
```

Shortly after that, the same IP requested a new file from the `/resumes/` directory:

```
2025-07-27 19:33:02 ... GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf ... 200
```

That showed me the uploaded resume had been stored by the application as:

```
Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf
```

```
2025-07-27 19:29:26 192.168.189.150 GET /resumes/WA_eCart - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 21
2025-07-27 19:29:26 192.168.189.150 GET /aspnet_client/cooking - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 20
2025-07-27 19:29:26 192.168.189.150 GET /resumes/bestselling - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 29
2025-07-27 19:29:26 192.168.189.150 GET /resumes/online - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 30
2025-07-27 19:29:26 192.168.189.150 GET /resumes/_xml - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 30
2025-07-27 19:29:26 192.168.189.150 GET /resumes/autos - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 30
2025-07-27 19:29:26 192.168.189.150 GET /resumes/instantlistings - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 32
2025-07-27 19:29:26 192.168.189.150 GET /resumes/basic - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 30
2025-07-27 19:29:26 192.168.189.150 GET /resumes/archived - 80 - 143.198.231.177 feroxbuster/2.11.0 - 404 0 2 30
2025-07-27 19:31:31 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 - 200 0 0 4
2025-07-27 19:32:18 192.168.189.150 GET /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 - 200 0 0 5
2025-07-27 19:32:28 192.168.189.150 POST /default.aspx - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/default.aspx 200 0 0 5
2025-07-27 19:33:02 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 - 200 0 0 11711
2025-07-27 19:33:17 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdirc=C:\inetpub\wwwroot\ - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\ 200 0 0 5
2025-07-27 19:33:23 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\ 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\inetpub\wwwroot\ 200 0 0 4
2025-07-27 19:33:28 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Users 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Users 200 0 0 5
2025-07-27 19:33:41 192.168.189.150 POST /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Users 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Users 200 0 0 5
2025-07-27 19:33:49 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\ 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Users 200 0 0 2
2025-07-27 19:33:55 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares 200 0 0 3
2025-07-27 19:34:02 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares\ResumeReview 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0
http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares 200 0 0 5
2025-07-27 19:34:23 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\Certificate_for_sign.cer 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/
20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares\ResumeReview 200 0 0 24
2025-07-27 19:34:35 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\idealcandidates.txt 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/
20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares\ResumeReview 200 0 0 3
2025-07-27 19:35:07 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\WhattoLookFor.txt 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/
20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares\ResumeReview 200 0 0 5
2025-07-27 19:35:13 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\ResumeReview.txt 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/
20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares\ResumeReview 200 0 0 3
2025-07-27 19:35:19 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares 200 0 0 3
2025-07-27 19:35:22 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\ 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdirc=C:\Shares 200 0 0 3
2025-07-27 19:35:51 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 - 200 0 0 5
2025-07-27 19:35:58 192.168.189.150 POST /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf 200 0 0 12
```

One thing I had to keep in mind from this point forward was URL encoding. Some IIS query strings used encodings like `%3a` for `:` and `%2f` for `/`, while spaces sometimes appeared as `+`. I decoded those values while writing the investigation so the Windows paths are easier to read.

Task 4: The threat actor uploaded another file in order to change the behavior of this website, allowing the first uploaded file to act as a webshell. What is the name of the uploaded file?

Answer: `web.config`

At first, the file looked like a normal PDF upload. The part that changed my direction was what happened next: the `.pdf` endpoint started accepting parameters such as `fdirc=`, `get=`, and `del=`.

Examples from the IIS log:

```
GET /resumes/Resume9_...pdf fdirc=C:\inetpub\wwwroot\
GET /resumes/Resume9_...pdf
get=C:\Shares\ResumeReview\Certificate_for_sign.cer
GET /resumes/Resume9_...pdf fdirc=C:\inetpub\wwwroot\resumes&del=...
```

A real PDF should not browse directories, retrieve files, or delete files. That meant IIS/ASP.NET was treating the uploaded PDF as server-side code.

This is where I had to research IIS behavior. Microsoft documents that IIS `<handlers>` mappings define how requests for specific extensions or URLs are processed, and that these mappings can be configured in `web.config` files. I also found Soroush Dalili's writeup on uploading `web.config` files, which shows how this can be abused in ASP.NET/IIS environments to make an uploaded file execute as server-side code.

Sources:

- [Microsoft IIS handlers documentation](#)
- [Microsoft ASP.NET Core web.config documentation](#)
- [Uploading web.config for Fun and Profit 2](#)

After that research, `web.config` made sense as the second uploaded file. It changed how the `/resumes/` directory handled the PDF file and allowed the first upload to behave as a web shell.

Task 5: Using the webshell, the threat actor uploaded a file on the endpoint to facilitate initial remote access. Identify the full path of this file.

Answer: `C:\inetpub\wwwroot\resumes\8619.exe`

Once I knew the PDF was acting as a web shell, I looked for tools placed in the same directory. The cleanup phase showed the attacker deleting this file:

```
2025-07-27 20:11:13 ... del=C:\inetpub\wwwroot\resumes\8619.exe ... 200
```

The USN Journal confirmed the file had been created earlier:

```
2025-07-27T19:35:58.859972+00:00 8619.exe DATA_EXTEND|FILE_CREATE|CLOSE
```

I also checked the filename/hash against public malware reputation sources during the original investigation. Those results were not the primary proof for the answer, but they helped me understand why `8619.exe` was likely an initial access payload instead of a normal uploaded file.

So the full path of the file used for initial remote access was:

```
C:\inetpub\wwwroot\resumes\8619.exe
```

```

#resmueupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf 200 0 0 12
Software: Microsoft Internet Information Services 10.0
versions: 1.0
Date: 2025-07-27 19:53:12
#fields: date time s-ip c-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) cs-status sc-win32-status time-taken
2025-07-27 20:11:16 192.168.189.150 GET /certify.exe 80 - 192.168.189.150 Mozilla/5.0+(Windows+X64;AppleWebKit/537.36+(KHTML,+likeGecko)/Chrome/136.0.0+Safari/537.36Fiddler/36.0.3240.50 - 404 0 2 16
2025-07-27 20:15:54 192.168.189.150 GET /resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\B691.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\Administrator.ccache 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\Administrator.ccache 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\B691.exe 200 0 6
2025-07-27 20:11:26 192.168.189.150 GET /resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\Administrator.ccache 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\Administrator.ccache 200 0 8
2025-07-27 20:11:32 192.168.189.150 GET /resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\Certify.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\Certify.exe 200 0 13
2025-07-27 20:11:34 192.168.189.150 GET /resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\PdfSigner.jsn 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\Certify.exe 200 0 6
2025-07-27 20:11:36 192.168.189.150 GET /resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\vr.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\vr.exe 200 0 6
2025-07-27 20:11:38 192.168.189.150 GET /resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\secretshdmp.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;Linux;x86_64rv:128.0)Gecko/20100101Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\secretshdmp.exe 200 0 6
2025-07-27 20:11:41 192.168.189.150 GET /resumes/Resume_eba15ba0-81ca-d0f9-ad-9fad-3fc1c92c181.pdf fdr:c:\inetpub\wwwroot\resumes\GedeC:\inetpub\wwwroot\resumes\vr.exe 200 0 6

```

JoeSandbox

Cloud

BASIC

Analyze

Results

Register

Login

Deep Malware Analysis

Details

Engines

IOCs

Joe Sandbox

Engine	Download Report	Detection	Info
JoeSandbox ML		MALICIOUS	
JoeSandbox Cloud BASIC Full Report Management Report IOC Report		MALICIOUS Score: 100	System: Windows 10 x64 22H2 with Office Professional Plus 2019, Chrome 134, Firefox...

Third Party Analysis Engines

VirusTotal

Open Full Report

MALICIOUS

Score: 25/71

REVERSINGLABS

MALICIOUS

Score: 8/41

57

/ 72

Community Score

57/72 security vendors flagged this file as malicious

80443f2c84ca76f98bbe9e84fe4a006b2b77b99eb94b0aa3b070197bb58d7

Fqon.exe

Size: 1.43 MB

Last Analysis Date: 1 year ago

EXE

peexe

detect-debug-environment

spreader

assembly

runtime-modules

direct-cpu-clock-access

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 5

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label

trojan.msil/loki

Threat categories

trojan

ransomware

Family labels

msil

loki

agenttesla

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Trojan/Win.PWSX-gen.C5272285	Alibaba	TrojanSpy:MSIL/AgentTesla.c993a3f3
AliCloud	Trojan[spy]:MSIL/AgentTesla.NG#2XJC	ALYac	Trojan.Ransom.Loki.COL
Arcabit	Trojan.Ransom.Loki.COL	Arctic Wolf	Unsafe
Avast	Win32:PWSX-gen [Trj]	AVG	Win32:PWSX-gen [Trj]
Avira (no cloud)	HEUR/AGEN.1307365	BitDefender	Trojan.Ransom.Loki.COL
Bkav Pro	W32.AIDetectMalware.CS	ClamAV	Win.Dropper.LokiBot-9974134-0
CrowdStrike Falcon	Win/malicious_confidence_100% (W)	CTX	Exe.trojan.msil
DeepInstinct	MALICIOUS	DrWeb	Trojan.MulDrop20.62568

Task 6: The threat actor accessed a shared folder meant for the resume reviewers. What is the full path of the first file they retrieved?

Answer: `C:\Shares\ResumeReview\Certificate_for_sign.cer`

The attacker used the web shell to browse from the web root to the root of `C:\`, then into `C:\Shares\ResumeReview`:

```
fdir=C:\
fdir=C:\Shares
fdir=C:\Shares\ResumeReview
```

The first `get=` request from that folder retrieved:

```
C:\Shares\ResumeReview\Certificate_for_sign.cer
```

After normalizing the double slash, the full path is:

```
C:\Shares\ResumeReview\Certificate_for_sign.cer
```

```

resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf 200 0 0 5
2025-07-27 19:33:23 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\ 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\inetpub\wwwroot\ 200 0 0 4
2025-07-27 19:33:28 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\Users 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 h
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\ 200 0 0 5
2025-07-27 19:33:41 192.168.189.150 POST /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\Users 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Users 200 0 0 916
2025-07-27 19:33:49 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\ 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Users 200 0 0 2
2025-07-27 19:33:55 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\Shares 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\ 200 0 0 3
2025-07-27 19:34:02 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\Shares\ResumeReview 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+
http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares 200 0 0 5
2025-07-27 19:34:23 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\Certificate_for_sign.cer 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares\ResumeReview 200 0 0 24
2025-07-27 19:34:35 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\ideal+candidates.txt 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares\ResumeReview 200 0 0 3
2025-07-27 19:35:07 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\What+to+look+for.txt 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares\ResumeReview 200 0 0 5
2025-07-27 19:35:13 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\ResumeReview.txt 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares\ResumeReview 200 0 0 3
2025-07-27 19:35:19 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\Shares 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares\ResumeReview 200 0 0 3
2025-07-27 19:35:20 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares 200 0 0 2
2025-07-27 19:35:22 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\ 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf?fdir=C:\Shares 200 0 0 3
2025-07-27 19:35:51 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 200 0 0 5
2025-07-27 19:35:58 192.168.189.150 POST /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf 200 0 0 12

```

Task 7: Who was the first ideal candidate the company was seeking?

Answer: warlocksmurf

While reviewing the IIS log, I saw that the attacker did not stop after retrieving the certificate file. They used the same PDF web shell to pull another file from the resume reviewer share:

```

2025-07-27 19:34:35 ... GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf get=C:\Shares\ResumeReview\ideal+candidates.txt ...
200

```

The + characters are URL encoding for spaces, so the retrieved file was:

```
C:\Shares\ResumeReview\ideal candidates.txt
```

At this point, I expected to simply open the file from the extracted evidence. That did not work because the original `C:\Shares\ResumeReview` folder and `ideal candidates.txt` were not directly available as normal files. So I had to treat it like a small file recovery problem.

The manual process would be:

1. Search the NTFS \$MFT for the filename `ideal candidates.txt`.
2. Find the matching file record.
3. Check whether the file's \$DATA attribute is resident or non-resident.
4. If it is resident, extract the file content directly from the MFT record.
5. Decode the recovered bytes as text.

The \$MFT, or Master File Table, stores metadata for files on an NTFS volume. For very small files, NTFS can store the actual file content inside the MFT record itself. This is called resident data. That matters here because the file content can still be recovered even when the original file is not present in the extracted folder.

Searching for `ideal candidates.txt` returned one matching record:

```


```

```
Record number: 913
Full path: /Shares/ResumeReview/ideal candidates.txt
Preferred name: ideal candidates.txt
State: In use
Type: File
```

Then I checked the **\$DATA** attribute:

```
$DATA attributes
Stream: $DATA
Resident: True
Size: 70
Retrieved resident content: 70 bytes
```

The important part is **Resident: True**. If this had been non-resident, I would have needed to follow NTFS data runs to the disk clusters that held the file content. Since it was resident, the 70 bytes of file content were stored directly inside MFT record **913**.

The recovered hex data was:

```
4c 6f 6f 6b 69 6e 67 20 66 6f 72 20 73 6f 6d 65
6f 6e 65 20 6c 69 6b 65 20 74 68 65 6d 0d 0a 2d
20 77 61 72 6c 6f 63 6b 73 6d 75 72 66 0d 0a 2d
20 74 6d 65 63 68 65 6e 0d 0a 2d 20 6c 75 6b 61
73 4c 55 5a 53 43
```

Decoding those bytes as text gave:

```
Looking for someone like them
- warlocksmurf
- tmechen
- lukasLUZSC
```

The first candidate listed was **warlocksmurf**.

The screenshot below makes the process look simple because I used my personal tool, [NTFS-Analyzer](#), to automate the repetitive MFT work. The tool lets me search for a filename, shows the matching record, identifies whether the **\$DATA** attribute is resident, and prints the recovered bytes as both hex and text. A reader can use that tool to reproduce the result instead of manually parsing the raw MFT structure byte by byte.

```

+ NTFS-Analyzer python3 mftparser.py -f /Users/hoangson/Documents/CTF/htb/JobApplicant/\$MFT -o output.csv --format csv --interactive
MFT filename lookup ready. Press Ctrl+D to exit.
File name> ideal candidates.txt
Found 1 match(es) for ideal candidates.txt.

Record Summary
Record number: 913
Header record number: 913
Sequence number: 8
Signature: FILE
State: In use
Type: File
Fixup OK: True
Full path: /Shares/ResumeReview/ideal candidates.txt
Preferred name: ideal candidates.txt
Base record:
Hard link count: 2
Used/allocated MFT bytes: 552/1024

$STANDARD_INFORMATION
Created: 2025-07-27T17:20:35.271188+00:00
Modified: 2025-07-27T17:23:52.489629+00:00
MFT modified: 2025-07-27T17:32:14.816789+00:00
Accessed: 2025-07-27T17:23:52.489629+00:00
Attributes: ARCHIVE

$FILE_NAME attributes
[1] Name: IDEALC-1.TXT
Namespace: DOS (2)
Parent record: 1112 seq 11
Created: 2025-07-27T17:20:35.271188+00:00
Modified: 2025-07-27T17:20:35.271188+00:00
MFT modified: 2025-07-27T17:20:35.271188+00:00
Accessed: 2025-07-27T17:20:35.271188+00:00
Real/allocated size: 0/0
Attributes: ARCHIVE
[2] Name: ideal candidates.txt
Namespace: WIN32 (1)
Parent record: 1112 seq 11
Created: 2025-07-27T17:20:35.271188+00:00
Modified: 2025-07-27T17:20:35.271188+00:00
MFT modified: 2025-07-27T17:20:35.271188+00:00
Accessed: 2025-07-27T17:20:35.271188+00:00
Real/allocated size: 0/0
Attributes: ARCHIVE

$SDATA attributes
[1] Stream: $DATA
Resident: True
Size: 70
Allocated size: 70
Initialized size: 70
Retrieved resident content: 70 bytes
Hex:
0000 4c 6f 6f 6b 69 6e 67 20 66 6f 72 20 73 6f 6d 65 Looking for some
0010 6f 6e 65 20 6c 69 6b 65 20 74 68 65 6d 0d 0a 2d one like them..-
0020 20 77 61 72 6c 6f 63 6b 73 6d 75 72 66 0d 0a 2d warlocksmurf..-
0030 20 74 6d 65 63 68 65 6e 0d 0a 2d 20 6c 75 6b 61 tmechen..- luka
0040 73 4c 55 5a 53 43 sLUZSC
Text:
Looking for someone like them
- warlocksmurf
- tmechen
- lukaLUZSC

```

Task 8: A file was dropped to perform a potential Kerberoasting attack in the Active Directory environment. What is the name of this file?

Answer: **r.exe**

I looked for attacker tools staged in the upload directory. During cleanup, the attacker deleted:

C:\inetpub\wwwroot\resumes\r.exe

The IIS cleanup request was:

2025-07-27 20:11:36 ... del=C:\inetpub\wwwroot\resumes\r.exe ... 200

The USN Journal showed that **r.exe** had been created much earlier:

2025-07-27T19:40:15.000068+00:00 r.exe DATA_EXTEND|FILE_CREATE|CLOSE

The short name fits the pattern of a renamed AD attack tool. In this timeline, it appears before the AD CS activity and later gets removed with the other attacker tools.

```
#Date: 2025-07-27 19:53:12
#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status time-taken
2025-07-27 19:53:12 192.168.189.150 GET /certsrv/ - 80 - 192.168.189.150 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/136.0.0.0+Safari/537.36+Edg/136.0.3240.50 - 404 0 2 16
2025-07-27 19:55:34 192.168.189.150 GET /certsrv/ - 80 - 192.168.189.150 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/136.0.0.0+Safari/537.36+Edg/136.0.3240.50 - 404 0 2 2
2025-07-27 20:10:46 192.168.189.150 POST /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://
resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf 200 0 0 2001179
2025-07-27 20:10:55 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 - 200 0 0 7
2025-07-27 20:11:13 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\8619.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf 200 0 0 8
2025-07-27 20:11:23 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\administrator.ccache 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\8619.exe
200 0 0 6
2025-07-27 20:11:26 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\8619.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\
administrator.ccache 200 0 0 8
2025-07-27 20:11:31 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\8619.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\
administrator.pfx 200 0 0 13
2025-07-27 20:11:34 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\8619.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\
PDFSigner.json 200 0 0 7
2025-07-27 20:11:36 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\8619.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\
PDFSigner.json 200 0 0 9
2025-07-27 20:11:38 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\secretsdump.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\
r.exe 200 0 0 9
2025-07-27 20:11:41 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\8619.exe 80 - 143.198.231.177 Mozilla/5.0+
(X11;+Linux;+x86_64;+rv:128.0)+Gecko/20100101+Firefox/128.0 http://resumeupload.wowza.edu/resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\
r.exe 200 0 0 9
```

Task 9: Which service account was targeted by this Active Directory attack?

Answer: `WOWZA\ca_svc`

For the AD portion, I moved from IIS logs to the Windows Security log. I first had to research which Windows events are useful for AD CS investigations. Microsoft's AD CS auditing references showed that Event ID 4886 is generated when Certificate Services receives a certificate request, and Event ID 4887 is generated when Certificate Services approves and issues one.

Sources:

- [Microsoft - Audit Certification Services](#)
- [Microsoft - Securing PKI: Events to Monitor](#)
- [Ultimate Windows Security - Event ID 4887](#)

With that context, Certificate Services Event ID 4886 showed a certificate request submitted by:

Requester: `WOWZA\ca_svc`

The same event contained suspicious request attributes:

CertificateTemplate:PDFSigner
SAN:upn=Administrator

So the attacker was using or abusing the `WOWZA\ca_svc` account to request certificates, while trying to place `Administrator` into the certificate's UPN.

Task 10: Attacker dropped another tool to exploit a misconfiguration in AD CS. When was this tool successfully created on the endpoint?

Answer: `2025-07-27 19:52:03`

The attacker later deleted **Certipy.exe** from the upload directory:

```
2025-07-27 20:11:31 ... del=C:\inetpub\wwwroot\resumes\\Certipy.exe ...
200
```

At first this was just another deleted executable, so I researched the filename before deciding why it mattered. Certipy is a public tool for Active Directory Certificate Services enumeration and abuse, and the SpecterOps "Certified Pre-Owned" research explains why AD CS misconfigurations can lead to privilege escalation and domain compromise.

Sources:

- [Certipy GitHub repository](#)
- [Certipy Wiki](#)
- [SpecterOps - Certified Pre-Owned](#)

To find when it was actually created, I checked the USN Journal. The first create event happened at **19:51:32**, but the important completed-write event was:

```
2025-07-27T19:52:03.015636+00:00 Certipy.exe
DATA_EXTEND|FILE_CREATE|CLOSE
```

The **CLOSE** flag showed that the file handle closed after the data was written. I treated that as the successful creation time.

```
(venv) + ntfs_parse git:(master) * python3 usnjrnlparse.py -f /Users/hoangson/Documents/CTF/htb/JobApplicant/\$Extend/\$J | grep Certipy.exe
88,2,0,20272016,2025-07-27 19:51:32.860094,FILE_CREATE,0,0,ARCHIVE,22,60,Certipy.exe,101786,9,101444,6
88,2,0,20272104,2025-07-27 19:51:43.843552,DATA_EXTEND|FILE_CREATE,0,0,ARCHIVE,22,60,Certipy.exe,101786,9,101444,6
88,2,0,20272496,2025-07-27 19:52:03.015636,DATA_EXTEND|FILE_CREATE|CLOSE,0,0,ARCHIVE,22,60,Certipy.exe,101786,9,101444,6
88,2,0,22746736,2025-07-27 20:11:31.386020,FILE_DELETE|CLOSE,0,0,ARCHIVE,22,60,Certipy.exe,101786,9,101444,6
```

Task 11: When was AD CS exploited successfully on the domain controller?

Answer: **2025-07-27 19:56:38**

I searched the Security log around the AD CS events and found the abuse beginning at:

```
2025-07-27T19:56:38.095869Z
```

The event was Security Event ID **4886**:

```
{
  "EventID": 4886,
  "RequestId": "6",
  "Requester": "WOWZA\\ca_svc",
```

```
"Attributes": "CertificateTemplate:PDFSigner\nSAN:upn=Administrator"
}
```

This was the first moment where the attacker requested a certificate from the vulnerable template while supplying **Administrator** as the UPN.

The same second also contained the template modification events that made the abuse possible. After checking Microsoft's PKI event list, I treated Event ID **4900** as especially important because it records certificate template security updates. Event ID **4900** showed the **PDFSigner** template security descriptor being changed, and Event ID **4898** showed the modified template content:

```
NewSecurityDescriptor: NT AUTHORITY\Authenticated Users - Full Control
msPKI-Certificate-Name-Flag: CT_FLAG_ENROLLEE_SUPPLIES_SUBJECT
pKIExtendedKeyUsage: Client Authentication
```

Those settings matter because they let an authenticated requester supply another identity in the certificate request and use the resulting certificate for authentication.

The later chain confirmed that the abuse worked:

- **19:56:49**: certificate issued for the suspicious request
- **19:59:15**: successful Kerberos TGT request for **Administrator** using certificate-based authentication

So the AD CS exploitation point was **2025-07-27 19:56:38**, with the successful authentication appearing a few minutes later.

Task 12: Identify the name and version of the vulnerable certificate template exploited in the attack.

Answer: **PDFSigner v100.4**

The certificate request pointed to the **PDFSigner** template, so I searched for template-load/change events involving that name.

I researched AD CS template events before trusting the field names. Microsoft lists Event ID **4898** as "Certificate Services loaded a template", and the event includes the template name, version, schema version, template content, and security descriptor. I also used Ultimate Windows Security and Lares Labs as supporting references for how this event appears during AD CS investigations.

Sources:

- [Microsoft - Audit Certification Services](#)
- [Ultimate Windows Security - Event ID 4898](#)
- [Lares Labs - AD CS Exploits & Investigations](#)

The relevant Security log entries showed:

```
TemplateInternalName: PDFSigner
TemplateVersion: 100.4
TemplateSchemaVersion: 2
```

That identified the exploited template as:

```
PDFSigner v100.4
```

The same template later appeared in the malicious request:

```
CertificateTemplate:PDFSigner
SAN:upn=Administrator
```

Task 13: A new security descriptor was applied to the vulnerable certificate template and this security descriptor grants full control over the certificate template to a specific group. What is the name of that group?

Answer: **Authenticated Users**

This part needed the certificate template change event, not just the request event. Security Event ID **4900** showed the **PDFSigner** template being changed at **2025-07-27T19:56:38.185611Z**.

This was a correction from my first pass. Event ID **4898** is useful for seeing the loaded template, but Event ID **4900** is the stronger evidence for this question because it records that the certificate template security was updated and includes the **NewSecurityDescriptor**.

The **NewSecurityDescriptor** contained:

```
Allow(0x000f01ff) NT AUTHORITY\Authenticated Users
Full Control
```

The short SID name in the descriptor was **AU**, which maps to **Authenticated Users**.

That was a dangerous change because **Authenticated Users** includes normal authenticated domain users. Giving that group Full Control over the certificate template made the template much easier to abuse.

```

pKIDefaultCSPs =
  Microsoft Enhanced Cryptographic Provider v1.0
  Microsoft Base Cryptographic Provider v1.0

msPKI-Supersede-Templates =

msPKI-RA-Policies =

msPKI-RA-Application-Policies =

msPKI-Certificate-Policy =

msPKI-Certificate-Application-Policy =
  1.3.6.1.5.5.7.3.2 Client Authentication

pKICriticalExtensions =
  2.5.29.19 Basic Constraints
  2.5.29.15 Key Usage
SecurityDescriptor:O:LAG:S-1-5-21-2525499130-77348690-3507557611-519D:PAI(A;;CCDCLCSWRPWPDTLOCRSDRCWDWO;;;AU)
  Allow(0x000f01ff)      NT AUTHORITY\Authenticated Users
                        Full Control

```

Task 14: When was the certificate issued to the threat actor for a high-privilege user, allowing privilege escalation?

Answer: 2025-07-27 19:57:55

I searched for another certificate request using the same pattern:

```

CertificateTemplate:PDFSigner
SAN:upn=Administrator

```

At 2025-07-27T19:57:55.575004Z, Event ID 4886 showed request ID 8:

```

{
  "EventID": 4886,
  "RequestId": "8",
  "Requester": "WOWZA\\ca_svc",
  "Attributes": "CertificateTemplate:PDFSigner\nSAN:upn=Administrator"
}

```

Immediately after that, Event ID 4887 showed the certificate was issued:

```

{
  "EventID": 4887,
  "SystemTime": "2025-07-27T19:57:55.676311Z",
  "RequestId": "8",
  "Requester": "WOWZA\\ca_svc",
  "Attributes": "CertificateTemplate:PDFSigner\nSAN:upn=Administrator",
  "Disposition": "3"
}

```

The matching **RequestId: 8** tied the issued certificate directly to the request for **Administrator**. The task asks for the issued time, so the rounded answer is:

2025-07-27 19:57:55

Task 15: The threat actor successfully retrieved the NTLM hash of a high-privilege user using the newly issued certificate. What was the serial number of the certificate used to achieve this privilege escalation?

Answer: 1D00000008A6E7524F029E603A000000000008

After the certificate was issued, I looked for Kerberos authentication using a certificate. Security Event ID **4768** showed a successful TGT request for **Administrator**:

```
{
  "EventID": 4768,
  "TargetUserName": "Administrator",
  "ServiceName": "krbtgt",
  "Status": "0x0",
  "PreAuthType": "16",
  "CertIssuerName": "wowza-MAIN-DC-CA",
  "CertSerialNumber": "1D00000008A6E7524F029E603A000000000008",
  "CertThumbprint": "15F5FE778E03BB9278B655827572EB2AA1AAFA9F"
}
```

The important parts were:

- **TargetUserName:** Administrator
- **Status:** 0x0, meaning success
- **PreAuthType:** 16, indicating certificate-based Kerberos authentication
- **CertSerialNumber**, which identifies the certificate used

I checked Microsoft's Event ID **4768** documentation while working through this because I wanted to be sure I was reading the Kerberos event correctly. Event ID **4768** is generated when a Kerberos TGT is requested, so a successful **4768** for **Administrator** after the certificate issuance connected the AD CS abuse to privileged authentication.

Source:

- [Microsoft - Event ID 4768: A Kerberos authentication ticket was requested](#)

Therefore, the serial number was:

1D00000008A6E7524F029E603A000000000008

Task 16: The attacker exploited an Active Directory misconfiguration to gain access to Active Directory secrets stored on the domain controller. What is the MITRE ATT&CK ID for this technique?

Answer: T1003.006

At this point in the timeline, the attacker had authenticated as Administrator. The next suspicious behavior was access to Active Directory replication permissions, which is the pattern used by DCSync.

DCSync lets an attacker with enough privilege ask the domain controller for replicated credential data, including password hashes.

I researched the technique name instead of guessing from the tool alone. MITRE ATT&CK maps DCSync to:

T1003.006 – OS Credential Dumping: DCSync

Source:

- [MITRE ATT&CK - OS Credential Dumping: DCSync \(T1003.006\)](#)

Task 17: When was the attack mentioned in the previous task successfully carried out?

Answer: 2025-07-27 20:05:14

To confirm DCSync, I looked for directory service access events with replication GUIDs.

The MITRE DCSync page gave me the technique name, but the Windows evidence still had to come from the Security log. That is why I focused on Event ID 4662 and replication-related GUIDs rather than just saying "secretsdump.exe existed".

First, the Security log showed an Administrator network logon:

```
{
  "EventID": 4624,
  "SystemTime": "2025-07-27T20:05:14.577498Z",
  "TargetUserName": "Administrator",
  "LogonType": 3,
  "IpAddress": "127.0.0.1",
  "TargetLogonId": "0x96a4b9"
}
```

Immediately after that, Event ID 4662 showed directory service access by the same logon session:

```
{
  "EventID": 4662,
  "SystemTime": "2025-07-27T20:05:14.687160Z",
  "SubjectUserName": "Administrator",
  "SubjectLogonId": "0x96a4b9",
  "ObjectServer": "DS",
  "AccessMask": "0x100"
}
```

The **Properties** field contained AD replication GUIDs:

```
{1131f6aa-9c07-11d1-f79f-00c04fc2dcd2}
{1131f6ad-9c07-11d1-f79f-00c04fc2dcd2}
```

Those correspond to replication permissions such as **DS-Replication-Get-Changes** and **DS-Replication-Get-Changes-All**, which are strong DCSync indicators.

The first DCSync-related event occurred in the **20:05:14** second, so the answer is:

```
2025-07-27 20:05:14
```

Task 18: A Golden Ticket was issued using a previously dropped tool. What is the filename of this ticket?

Answer:

shinyboi_wowza_edu_2025_07_27_20_06_33_Administratorr_to_krbtgt@WOWZA.EDU.kirbi

After DCSync, I expected to see Kerberos ticket artifacts. The USN Journal showed a **.kirbi** file being created:

```
2025-07-27T20:06:33.604956+00:00
shinyboi_wowza_edu_2025_07_27_20_06_33_Administratorr_to_krbtgt@WOWZA.ED
U.kirbi DATA_EXTEND|FILE_CREATE|CLOSE
```

The same filename appeared later during cleanup in the IIS log. In the log it was URL-encoded as **%40**, which decodes to **@**:

```
shinyboi_wowza_edu_2025_07_27_20_06_33_Administratorr_to_krbtgt%40WOWZA.
EDU.kirbi
```

Decoded, the filename is:

```
shinyboi_wowza_edu_2025_07_27_20_06_33_Administratorr_to_krbtgt@WOWZA.EDU.kirbi
```

The **krbtgt** part of the filename fits a Golden Ticket because Golden Tickets are forged Kerberos TGTs tied to the **krbtgt** account. I checked this against Microsoft Kerberos references and MITRE's Golden Ticket technique page before labeling it that way. Microsoft documents that the KDC uses the **krbtgt** security principal for ticket-granting activity, and MITRE describes Golden Tickets as forged Kerberos TGTs created using the **KRBTGT** account password hash.

Sources:

- [Microsoft - Key Distribution Center](#)
- [Microsoft - Ticket-Granting Tickets](#)
- [MITRE ATT&CK - Golden Ticket \(T1558.001\)](#)

Task 19: The threat actor used the web shell to delete all files involved in this operation. When did the deletion routine begin?

Answer: 2025-07-27 20:11:13

The cleanup routine was visible in the IIS log because the attacker used the web shell with the **del=** parameter.

The first deletion request was:

```
2025-07-27 20:11:13 ... del=C:\inetpub\wwwroot\resumes\8619.exe ... 200
```

The USN Journal confirmed the matching delete event:

```
2025-07-27T20:11:13.980332+00:00 8619.exe FILE_DELETE|CLOSE
```

After that, the attacker deleted the rest of the operation files:

- administrator.ccache
- administrator.pfx
- Certipy.exe
- PDFSigner.json
- r.exe
- secretsdump.exe
- shinyboi_wowza_edu_2025_07_27_20_06_33_Administratorr_to_krbtgt@WOWZA.EDU.kirbi

So the deletion routine began at:

2025-07-27 20:11:13

```
2025-07-27 19:33:22 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101Firefox/12.0
2025-07-27 19:35:51 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101Firefox/12.0
2025-07-27 19:35:58 192.168.189.150 POST /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101Firefox/12.0
#Software: Microsoft Internet Information Services 10.0
#Version: 1.0
#Date: 2025-07-27 19:53:12
#Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status time-taken
2025-07-27 19:53:12 192.168.189.150 GET /certsrv/ - 80 - 192.168.189.150 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/136.0.0.0+Safari/537.36+Edg
2025-07-27 19:55:34 192.168.189.150 GET /certsrv/ - 80 - 192.168.189.150 Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64)+AppleWebKit/537.36+(KHTML,+like+Gecko)+Chrome/136.0.0.0+Safari/537.36+Edg
2025-07-27 20:10:46 192.168.189.150 POST /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101Firefox/12.0
2025-07-27 20:10:55 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf - 80 - 143.198.231.177 Mozilla/5.0+(X11;Linux;x86_64;rv:128.0)Gecko/20100101Firefox/12.0
2025-07-27 20:11:13 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\8619.exe 80 - 143.198.231.177
2025-07-27 20:11:23 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\administrator.ccache 80 - 143.198.231.177
2025-07-27 20:11:26 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\administrator.pfx 80 - 143.198.231.177
2025-07-27 20:11:31 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\Certipy.exe 80 - 143.198.231.177
2025-07-27 20:11:34 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\PDFSigner.json 80 - 143.198.231.177
2025-07-27 20:11:36 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\r.exe 80 - 143.198.231.177
2025-07-27 20:11:38 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\secretsdump.exe 80 - 143.198.231.177
2025-07-27 20:11:41 192.168.189.150 GET /resumes/Resume9_eba15ba0-81ca-4d0f-9fad-3fc1fc92c181.pdf fdir=C:\inetpub\wwwroot\resumes\&del=C:\inetpub\wwwroot\resumes\shinyboi_wowza_edu_2025_0
```

Timeline Summary

2025-07-27 19:27:35	First suspicious web request from 143.198.231.177
2025-07-27 19:32:28	Resume upload request
2025-07-27 19:33:02	Stored PDF accessed
2025-07-27 19:33:17	Uploaded PDF begins behaving as a web shell
2025-07-27 19:35:58	8619.exe successfully created
2025-07-27 19:40:15	r.exe successfully created
2025-07-27 19:52:03	Certipy.exe successfully created
2025-07-27 19:56:38	Malicious AD CS certificate request/template abuse
2025-07-27 19:57:55	Certificate issued for SAN:upn=Administrator
2025-07-27 19:59:15	Certificate-based TGT request for Administrator
2025-07-27 20:05:14	DCSync activity begins
2025-07-27 20:06:33	Golden Ticket .kirbi file created
2025-07-27 20:11:13	Cleanup/deletion routine begins

References

These are the outside references I used while turning log clues into explanations:

- [Microsoft IIS handlers documentation](#)
- [Microsoft ASP.NET Core web.config documentation](#)
- [Uploading web.config for Fun and Profit 2](#)
- [NTFS-Analyzer](#)
- [Microsoft - Audit Certification Services](#)
- [Microsoft - Securing PKI: Events to Monitor](#)
- [Ultimate Windows Security - Event ID 4887](#)
- [Certipy GitHub repository](#)
- [Certipy Wiki](#)
- [SpecterOps - Certified Pre-Owned](#)
- [Ultimate Windows Security - Event ID 4898](#)

- [Lares Labs - ADCS Exploits & Investigations](#)
- [Microsoft - Event ID 4768: A Kerberos authentication ticket was requested](#)
- [MITRE ATT&CK - OS Credential Dumping: DCSync \(T1003.006\)](#)
- [Microsoft - Key Distribution Center](#)
- [Microsoft - Ticket-Granting Tickets](#)
- [Microsoft - Kerberos Policy](#)
- [MITRE ATT&CK - Golden Ticket \(T1558.001\)](#)